

Lab 01:

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1  -- Window function lab
2  -- Lab 01:
3
4  WITH RankedFilms AS(
5      SELECT c.name AS category, f.title AS film_title, f.length AS film_len,
6             DENSE_RANK() OVER(PARTITION BY c.name ORDER BY f.length DESC, f.title) AS rn
7      FROM film f
8      JOIN film_category fc ON f.film_id = fc.film_id
9      JOIN category c ON c.category_id = fc.category_id
10 )
11
12 SELECT * FROM RankedFilms WHERE rn <= 3
```

	category character varying (25)	film_title character varying (255)	film_len smallint	rn bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3

Lab 02:

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-- Lab 02:
WITH monthly_revenue AS (
    SELECT
        DATE_TRUNC('month', payment_date) AS revenue_month,
        SUM(amount) AS total_revenue
    FROM payment
    GROUP BY 1
),
lagged_revenue AS (
    SELECT revenue_month, total_revenue,
        LAG(total_revenue, 1) OVER (
            ORDER BY revenue_month ASC
        ) AS previous_month_revenue
    FROM monthly_revenue
)
SELECT revenue_month, total_revenue, previous_month_revenue,
    ROUND(
        (total_revenue - previous_month_revenue) * 100.0 / previous_month_revenue, 2
    ) AS mom_growth_pct
FROM lagged_revenue;
```

Lab 03:

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36 -- Lab 03:
37 WITH payment_details AS (
38     SELECT payment_id, customer_id, payment_date, amount
39     FROM payment
40 )
41 SELECT payment_id, customer_id, payment_date, amount,
42     SUM(amount) OVER (
43         PARTITION BY customer_id
44         ORDER BY payment_date ASC
45     ) AS running_total_clv
46 FROM payment_details;
```

Data Output Messages Notifications

	payment_id [PK] integer	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	running_total_clv numeric
1	18495	1	2007-02-14 23:22:38.996577	5.99	5.99
2	18496	1	2007-02-15 16:31:19.996577	0.99	6.98
3	18497	1	2007-02-15 19:37:12.996577	9.99	16.97
4	18498	1	2007-02-16 13:47:23.996577	4.99	21.96
5	18499	1	2007-02-18 07:10:14.996577	4.99	26.95
6	18500	1	2007-02-18 12:02:25.996577	0.99	27.94
7	18501	1	2007-02-21 04:53:11.996577	3.99	31.93
8	22680	1	2007-03-01 07:19:30.996577	4.99	36.92
9	22681	1	2007-03-02 14:05:18.996577	3.99	40.91
10	22682	1	2007-03-02 16:30:04.996577	0.99	41.90
11	22683	1	2007-03-17 11:06:20.996577	4.99	46.89
12	22684	1	2007-03-18 02:25:55.996577	0.99	47.88

Lab 04:

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48 -- Lab 04:
49 WITH customer_spending AS (
50     SELECT co.country AS country_name, c.customer_id, c.first_name || ' ' || c.last_name AS customer_name,
51           SUM(p.amount) AS total_amount
52     FROM customer c
53     JOIN address a ON c.address_id = a.address_id
54     JOIN city ci ON a.city_id = ci.city_id
55     JOIN country co ON ci.country_id = co.country_id
56     JOIN payment p ON c.customer_id = p.customer_id
57     GROUP BY co.country, c.customer_id, c.first_name, c.last_name
58 ),
59 ranked_customers AS (
60     SELECT country_name, customer_id, customer_name, total_amount,
61           RANK() OVER (
62             PARTITION BY country_name
63             ORDER BY total_amount DESC
64           ) AS customer_rank
65     FROM customer_spending
66 )
67 SELECT country_name, customer_id, customer_name, total_amount
68 FROM ranked_customers
69 WHERE customer_rank = 1;

```

	country_name character varying (50)	customer_id integer	customer_name text	total_amount numeric
1	Afghanistan	218	Vera Mccoy	67.82
2	Algeria	176	June Carroll	151.68
3	American Samoa	320	Anthony Schwab	47.85
4	Angola	528	Claude Herzog	93.80
5	Anguilla	381	Bobby Boudreau	99.68
6	Argentina	560	Jordan Archuleta	129.71
7	Armenia	41	Stephanie Mitch...	118.75
8	Austria	266	Nora Herrera	113.74
9	Azerbaijan	334	Raymond Mcwh...	108.75
10	Bahrain	590	Seth Hannon	108.76
11	Bangladesh	21	Michelle Clark	146.68
12	Belarus	144	Clara Shaw	189.60
13	Bolivia	431	Joel Francisco	92.78
14	Brazil	178	Marion Snyder	194.61
15	Brunei	91	Lois Butler	107.66
16	Bulgaria	540	Tyrone Asher	112.76
17	Cambodia	516	Elmer Noe	92.76
18	Cameroon	352	Albert Crouse	96.78
19	Canada	410	Curtis Irby	167.62
20	Chad	535	Javier Elrod	122.72
21	Chile	45	Janet Phillips	124.74
22	China	550	Guy Brownlee	151.69
23	Colombia	87	Wanda Patterson	137.72